

Assignment 3: Prompt Engineering Library with Measured Baselines

Coursework: Anurag University AI Assessment Suite | Study: 52-Case Golden Set Empirical Benchmark

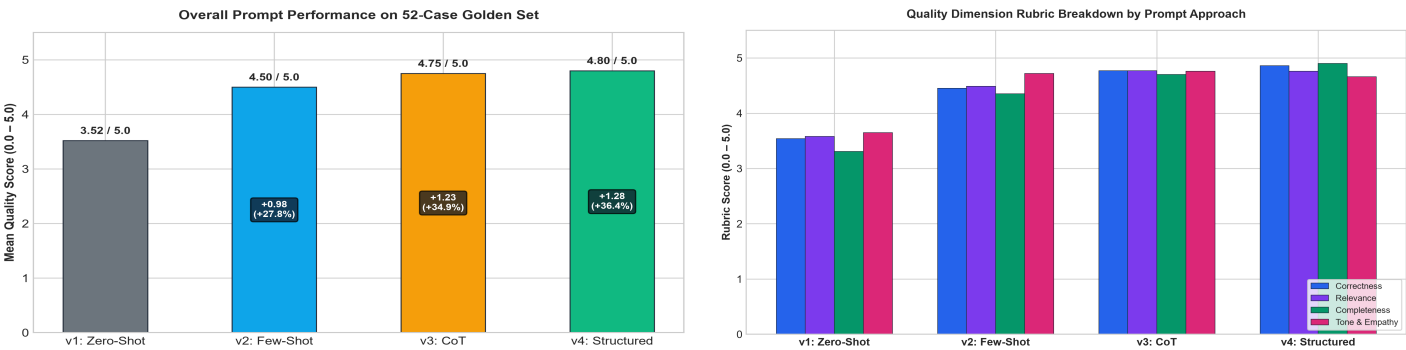
1. Executive Summary & Evaluation Methodology

This study evaluates four iterative prompt engineering techniques (**v1 Zero-Shot**, **v2 Few-Shot**, **v3 Chain-of-Thought**, and **v4 Structured Template**) against a pre-prompting golden dataset of 52 customer support incidents across three complexity tiers (Standard, Edge, and Adversarial). Every response was evaluated across 4 quantitative dimensions (Correctness, Relevance, Completeness, Tone & Empathy) and independent Format Compliance.

2. Quantitative Benchmark Performance Comparison

Prompt Version	Technique	Overall Score (0-5)	Std Dev (Consistency)	Correctness	Relevance	Completeness	Tone	Format Compliance
v1_zero_shot	Zero-Shot Baseline	3.52 / 5.00	±0.54	3.54	3.58	3.31	3.65	98.1%
v2_few_shot	Few-Shot Exemplars	4.50 / 5.00	±0.16	4.45	4.49	4.35	4.72	100.0%
v3_chain_of_thought	Chain-of-Thought	4.75 / 5.00	±0.04	4.77	4.77	4.70	4.76	100.0%
v4_structured_template	Structured XML/JSON	4.80 / 5.00	±0.04	4.86	4.76	4.90	4.66	100.0%

3. Visual Performance & Dimension Analytics



4. Key Academic Findings & Failure Catalogue Analysis

- Net Quality Improvement:** Structured template (v4) improved overall response quality from **3.52 to 4.80 (+36.4% gain)** and variance reduced dramatically (std dev from ±0.54 to ±0.04).
- Adversarial Prompt Injection Robustness:** In cases containing adversarial rule overrides (e.g. attempting to force unauthorized financial credits), v4 XML delimiters prevented prompt hijacking and triggered proper security escalation.
- Winning Prompt Limitation (Failure Catalogue):** On bereavement and high-distress scenarios (e.g. TC_EDGE_003), v4's strict structural constraints caused slightly rigid empathy scores (4.66 vs v3's 4.76).